



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal TO A.C.M.PETROLEUM AGENCIES

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

## Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

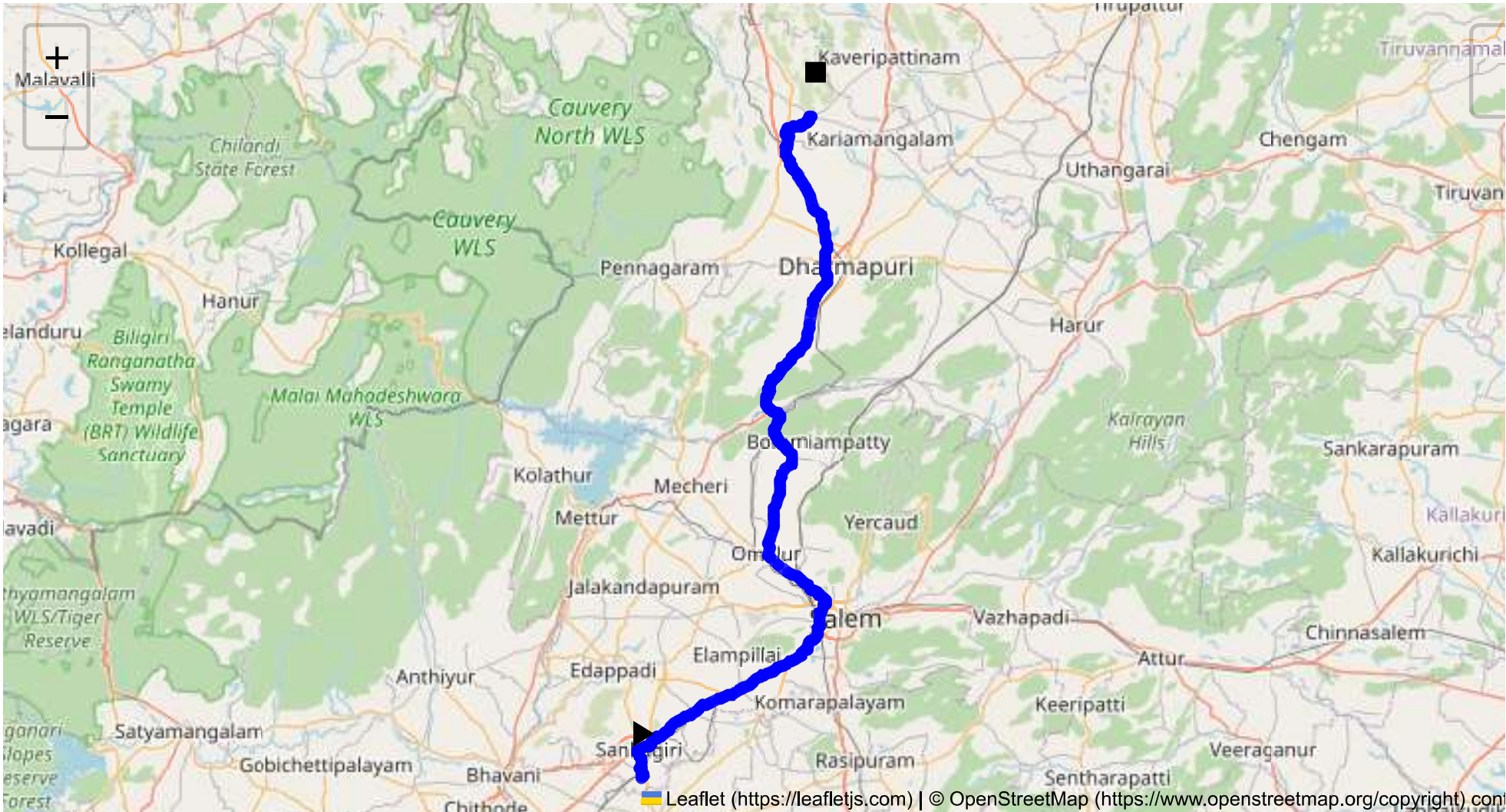
## Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.





**Route Summary:**  
**Total Distance: 132.35 km**  
**Estimated Duration: 2.5 hours**  
**Adjusted Duration (Heavy Vehicle): 3.1 hours**  
**Start: (11.4381, 77.8734)**  
**End: (12.33811, 78.11145)**



## Welcome to the Journey Risk Management Study

### 1. Overview of the Route Map:

- The route from CVQF+23W, Sangagiri to 84Q6+6H, Doddaradhanahalli covers approximately 132.35 kilometers. The standard path likely follows major highways and arterial roads to ensure smooth travel for heavy vehicles. It's pertinent to note important junctions and interchanges, such as potential connectivity with NH44 and state highways.



## 2. Typical Weather Conditions and Potential Weather-Related Hazards:

- Tamil Nadu typically experiences a tropical climate with hot summers, moderate monsoons, and mild winters. During monsoon season, the route can face heavy rainfall, leading to waterlogging, slick roads, and reduced visibility. Summers can cause road surface issues due to high temperatures, which may lead to tar melting in certain areas, affecting traction.

## 3. Traffic Patterns:

- Traffic congestion likely peaks during morning (8-10 AM) and evening (5-8 PM) on weekdays, particularly in areas like town centers or highway junctions. Doddaradhanahalli and Sangagiri might have local congestion due to local traffic from markets or educational institutions.

## 4. Road Quality and Infrastructure:

- While national and state highways are generally well-maintained, ensuring smooth and broad lanes, rural segments or interconnecting roads might be prone to potholes, inadequate signage, or absence of proper lighting during night hours. Bridges and narrow passes can be pinch points for heavy trucks.

## 5. Suggestions for Alternative Routes:

- In cases of significant blockage or emergencies, utilizing state highways as interim routes is advisable. Planning a detour through less congested rural roads might be beneficial, but only if they are reported to be in good condition and are wide enough to support heavy vehicles without risk.

## 6. Local Regulations Affecting Hazardous Material Transport:

- Transporting hazardous materials mandates strict adherence to central and state transport regulations, including proper documentation, vehicle signages, and possibly requiring special permits through certain ecologically sensitive areas.

## 7. Historical Incidents:

- Check local databases or traffic reports for any history of accidents involving heavy vehicles. Awareness of recent accidents, particularly those involving hazardous materials, can illuminate high-risk areas or validate safer alternatives.

## 8. Environmental Considerations and Sensitive Areas:

- Environmental sensitivity might involve passing near water bodies or forested areas. It's crucial to prevent spillage and follow guidelines for noise pollution reduction to protect wildlife and aquatic life.

## 9. Communication Coverage:

- The route should have overall good network coverage, especially along highways. However, rural stretches might have dead zones. Drivers should plan to enter potentially uncovered areas with all necessary information saved offline or have a satellite communication backup if feasible.

## 10. Estimated Emergency Response Times:

- Response times can vary, with urban areas along the route generally quicker due to proximity to facilities. Rural segments could experience delays, often exceeding 30 minutes, based on the distance from the nearest major town or city with emergency services.

11. Overall Summary of Risk Assessment:

- The route presents a moderate level of risk primarily due to weather conditions during certain seasons and traffic congestion at peak hours. Road conditions vary, with highways offering reliable quality but attention needed for rural links. Adherence to legal transport guidelines and constant communication readiness are crucial for mitigating risks associated with transporting hazardous materials. Regular updates on traffic and weather conditions are recommended to ensure maximum safety.

Risk Assessment - Turns

|    | Risk Type | Risk Level | Coordinates        | Speed Limit |
|----|-----------|------------|--------------------|-------------|
| 0  | Turn      | High       | 11.43968, 77.87345 | 15 KM/Hr    |
| 1  | Turn      | High       | 11.44029, 77.87544 | 15 KM/Hr    |
| 2  | Turn      | High       | 11.44900, 77.87406 | 15 KM/Hr    |
| 3  | Turn      | High       | 11.45028, 77.87202 | 15 KM/Hr    |
| 4  | Turn      | Medium     | 11.45309, 77.87325 | 30 KM/Hr    |
| 5  | Turn      | Medium     | 11.46400, 77.87323 | 30 KM/Hr    |
| 6  | Turn      | Medium     | 11.46765, 77.87037 | 30 KM/Hr    |
| 7  | Turn      | High       | 11.47403, 77.86841 | 15 KM/Hr    |
| 8  | Turn      | High       | 11.49149, 77.89601 | 15 KM/Hr    |
| 9  | Turn      | High       | 11.49224, 77.89601 | 15 KM/Hr    |
| 10 | Turn      | Medium     | 11.53501, 77.95707 | 30 KM/Hr    |
| 11 | Turn      | Medium     | 11.53601, 77.95856 | 30 KM/Hr    |
| 12 | Turn      | Medium     | 11.62403, 78.10893 | 30 KM/Hr    |
| 13 | Turn      | Medium     | 11.64180, 78.11808 | 30 KM/Hr    |
| 14 | Turn      | Medium     | 11.64325, 78.11993 | 30 KM/Hr    |
| 15 | Turn      | Medium     | 11.69519, 78.10775 | 30 KM/Hr    |
| 16 | Turn      | Medium     | 11.73958, 78.05105 | 30 KM/Hr    |
| 17 | Turn      | Medium     | 11.75835, 78.05016 | 30 KM/Hr    |
| 18 | Turn      | Medium     | 11.87942, 78.08197 | 30 KM/Hr    |
| 19 | Turn      | Medium     | 11.92585, 78.06507 | 30 KM/Hr    |
| 20 | Turn      | Medium     | 11.93704, 78.05402 | 30 KM/Hr    |
| 21 | Turn      | Medium     | 11.95818, 78.04978 | 30 KM/Hr    |
| 22 | Turn      | Medium     | 11.95876, 78.04990 | 30 KM/Hr    |

|    | Risk Type | Risk Level | Coordinates        | Speed Limit |
|----|-----------|------------|--------------------|-------------|
| 23 | Turn      | Medium     | 11.95988, 78.04948 | 30 KM/Hr    |
| 24 | Turn      | Medium     | 11.96304, 78.05136 | 30 KM/Hr    |
| 25 | Turn      | High       | 11.96399, 78.05135 | 15 KM/Hr    |
| 26 | Turn      | High       | 11.96580, 78.04923 | 15 KM/Hr    |
| 27 | Turn      | Medium     | 11.96675, 78.04939 | 30 KM/Hr    |
| 28 | Turn      | Medium     | 11.96791, 78.05070 | 30 KM/Hr    |
| 29 | Turn      | Medium     | 12.12058, 78.13283 | 30 KM/Hr    |
| 30 | Turn      | High       | 12.26533, 78.08428 | 15 KM/Hr    |
| 31 | Turn      | High       | 12.26550, 78.08440 | 15 KM/Hr    |
| 32 | Turn      | Medium     | 12.27173, 78.08098 | 30 KM/Hr    |
| 33 | Turn      | High       | 12.29595, 78.07368 | 15 KM/Hr    |
| 34 | Turn      | Medium     | 12.29688, 78.07580 | 30 KM/Hr    |
| 35 | Turn      | Medium     | 12.31153, 78.07860 | 30 KM/Hr    |
| 36 | Turn      | Medium     | 12.31236, 78.07799 | 30 KM/Hr    |
| 37 | Turn      | High       | 12.31360, 78.07464 | 15 KM/Hr    |
| 38 | Turn      | High       | 12.32018, 78.07656 | 15 KM/Hr    |
| 39 | Turn      | Medium     | 12.32065, 78.08368 | 30 KM/Hr    |
| 40 | Turn      | Medium     | 12.32338, 78.08719 | 30 KM/Hr    |
| 41 | Turn      | High       | 12.32321, 78.09514 | 15 KM/Hr    |
| 42 | Turn      | Medium     | 12.33151, 78.10541 | 30 KM/Hr    |
| 43 | Turn      | Medium     | 12.33697, 78.10836 | 30 KM/Hr    |

Emergency Locations

|   | type     | name                    | coordinates               | speed_limit | risk_level |
|---|----------|-------------------------|---------------------------|-------------|------------|
| 0 | hospital | Thilagam Hospital       | 11.4753573,<br>77.8685829 | 30 km/h     | Medium     |
| 1 | clinic   | Meyyazhagan Clinic      | 11.4752254,<br>77.8685671 | 30 km/h     | Medium     |
| 2 | hospital | Dr. Jaganathan Hospital | 11.4752717,<br>77.8709312 | 30 km/h     | Medium     |
| 3 | hospital | Kathir Eye Hospital     | 11.4740976,<br>77.8681591 | 30 km/h     | Medium     |

|    | type     | name                                                            | coordinates               | speed_limit | risk_level |
|----|----------|-----------------------------------------------------------------|---------------------------|-------------|------------|
| 4  | police   | Police Quarters                                                 | 11.4741243,<br>77.8673495 | 30 km/h     | Medium     |
| 5  | hospital | Sankari Hospital                                                | 11.4777496,<br>77.873892  | 30 km/h     | Medium     |
| 6  | clinic   | Dr. Dhanasekar Clinic                                           | 11.4793113,<br>77.8829247 | 30 km/h     | Medium     |
| 7  | hospital | Sri Vellaiyan Hospital                                          | 11.4798876,<br>77.8831394 | 30 km/h     | Medium     |
| 9  | hospital | C. N. Hospital                                                  | 11.55724, 78.0096         | 30 km/h     | Medium     |
| 10 | hospital | Vinayaka Mission's Homeopathic Medical College & Hospital       | 11.583617,<br>78.059145   | 30 km/h     | Medium     |
| 11 | hospital | Vinayaka mission Kirubananadah variyar medical college hospital | 11.584984,<br>78.0618417  | 30 km/h     | Medium     |
| 12 | hospital | vims                                                            | 11.5854239,<br>78.0639188 | 30 km/h     | Medium     |
| 15 | hospital | Vinayaka Mission’s Kirupananda Variyar Medical College          | 11.5977669,<br>78.0850992 | 30 km/h     | Medium     |
| 19 | hospital | Bharathi Hospital, Salem                                        | 11.6492647,<br>78.1204131 | 30 km/h     | Medium     |
| 20 | hospital | Sisu Hospital Inida Ltd                                         | 11.6505276,<br>78.1205003 | 30 km/h     | Medium     |
| 21 | hospital | Aishwarya Hospital                                              | 11.6716291,<br>78.1267807 | 30 km/h     | Medium     |
| 22 | hospital | C.M. Hospital                                                   | 11.6747345,<br>78.1255034 | 30 km/h     | Medium     |
| 24 | pharmacy | Healthy Pharmacy                                                | 11.6762714,<br>78.127744  | 30 km/h     | Medium     |
| 25 | pharmacy | Thulasi Pharmacy                                                | 11.6761991,<br>78.1277377 | 30 km/h     | Medium     |
| 26 | hospital | Sri Gokulam Speciality Hospital                                 | 11.6788724,<br>78.1274258 | 30 km/h     | Medium     |
| 27 | hospital | Global Medical Center                                           | 11.6867049,<br>78.1186697 | 30 km/h     | Medium     |
| 28 | hospital | Manipal Hospital                                                | 11.703072,<br>78.1026612  | 30 km/h     | Medium     |
| 31 | clinic   | Manimuthu Skin Clinic                                           | 11.7521189,<br>78.0484058 | 30 km/h     | Medium     |
| 32 | hospital | St. Joseph's Speciality Hospital                                | 11.761611,<br>78.0522344  | 30 km/h     | Medium     |

|    | type     | name                               | coordinates               | speed_limit | risk_level |
|----|----------|------------------------------------|---------------------------|-------------|------------|
| 33 | hospital | Anugraha Multi Speciality Hospital | 11.773549,<br>78.0546408  | 30 km/h     | Medium     |
| 34 | clinic   | JSA Clinic                         | 11.8700619,<br>78.0844982 | 30 km/h     | Medium     |
| 35 | hospital | L.R. Hospital                      | 11.8703354,<br>78.0844554 | 30 km/h     | Medium     |
| 37 | hospital | Government Hospital Palacode       | 12.2964824,<br>78.0761735 | 30 km/h     | Medium     |

Crowded Spots

|    | type       | name                                                    | coordinates               | speed_limit | risk_level |
|----|------------|---------------------------------------------------------|---------------------------|-------------|------------|
| 13 | university | Vinayaka Missions Vinayaka Missions Research Foundation | 11.5974056,<br>78.0843617 | 30 km/h     | Medium     |
| 14 | school     | Vinayaka Missions Annapoorani Matric School             | 11.5971662,<br>78.0839876 | 30 km/h     | Medium     |
| 16 | school     | Dist Education Training                                 | 11.6133799,<br>78.1022409 | 30 km/h     | Medium     |
| 17 | school     | DIET - Salem                                            | 11.6144158,<br>78.1025078 | 30 km/h     | Medium     |
| 18 | school     | Diamond Rays Matric HR Sec School                       | 11.6385629,<br>78.1206259 | 30 km/h     | Medium     |
| 23 | college    | Sona College                                            | 11.6768,<br>78.1257052    | 30 km/h     | Medium     |
| 29 | university | Periyar University                                      | 11.7186758,<br>78.0771089 | 30 km/h     | Medium     |
| 30 | school     | Fathima Hr Sec School                                   | 11.7410508,<br>78.0479272 | 30 km/h     | Medium     |
| 36 | college    | Laxminarayana Polytechnic College                       | 12.0421277,<br>78.103203  | 30 km/h     | Medium     |

Route Photos of Risky Spots





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.44029, 77.87544



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.44900, 77.87406





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.45028, 77.87202



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45309, 77.87325



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.46400, 77.87323





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.46765, 77.87037



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.47403, 77.86841





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.49149, 77.89601



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.49224, 77.89601



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.53501, 77.95707





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.53601, 77.95856



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.62403, 78.10893





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.64180, 78.11808



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.64325, 78.11993



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.69519, 78.10775





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.73958, 78.05105



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.75835, 78.05016





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.87942, 78.08197



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.92585, 78.06507



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.93704, 78.05402





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.95818, 78.04978



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.95876, 78.04990





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.95988, 78.04948



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.96304, 78.05136



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.96399, 78.05135





**Risk Type: Turn**

**Risk Level: High**

**Speed Limit: 15 KM/Hr**

**Coordinates: 11.96580, 78.04923**



**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.96675, 78.04939**





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.96791, 78.05070



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.12058, 78.13283



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.26533, 78.08428





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.26550, 78.08440



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.27173, 78.08098





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.29595, 78.07368



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.29688, 78.07580



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.31153, 78.07860





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.31236, 78.07799



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.31360, 78.07464





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.32018, 78.07656



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.32065, 78.08368



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.32338, 78.08719





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.32321, 78.09514



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.33151, 78.10541





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 12.33697, 78.10836

